

Literature Review: Base Papers on Phishing URL Detection

Extracted directly from each paper’s dataset, methodology, results, and limitations sections

Paper	Dataset Used	Size	Phishing URLs	Legitimate URLs	Features Used	Algorithm	Evaluation Metrics	Accuracy
ResMLP (Remya et al., IEEE Access 2024)	Kaggle dataset (via PhishHaven ref.), filtered to benign+phishing	522,214 (filtered; 651,191 raw)	94,111	428,103	25 URL-derived lexical features; WHOIS domain age; Google-index status; GLOVE+NLTK sentiment score	ResMLP: conv block + 7 inverted-residual blocks + dense/softmax	Accuracy, Precision, Recall, F1, ROC-AUC	98.29%
Feature Extension (He et al., IEEE IoT J. 2024)	PhishTank (phishing) + Kaggle “Phishing site URLs” (legit), balanced	73,576	36,788	36,788	TextRank feature-extension library (top-10 keywords) + BERT WordPiece embedding (vector length 138)	Custom BERT (6 layers, H=256) → 1D-CNN classifier	Accuracy, Precision, Recall, F1, FPR, AUC	99.75%
Multimodal XAI (Vulfin et al., MAKE 2026)	Proprietary D1 + public MTLP (D2)	D1 ≈3.2M (cleaned); D2 = 15,000	D1: 1.6M; D2: ≈50% (not printed)	D1: 1.6M; D2: ≈50% (not printed)	URL+metadata (>89 feats incl. WHOIS/SSL); char-level URL (CNN1D); HTML (CodeBERT); screenshot (EfficientNet-B7)	Late-fusion ensemble: CatBoost + CNN1D + CodeBERT + EfficientNet-B7 + meta-classifier	Accuracy, Precision, Recall, F1, ROC-AUC, FPR, FNR	F1 0.989 (D1) / 0.953 (D2)
Evaluating Feature Engineering (Kustiawan & Ghauth, IEEE Access 2025)	PhishOFE dataset	101,063	48.97% (≈49,491)	51.03% (≈51,572)	URL (11 feats) + HTML (19 feats) + Derived (4 feats), tested in 4 combinations	10 ML models compared (RF, DT, kNN, GBM, LR, LightGBM, SVM, CatBoost, NB, XGBoost); best = CatBoost	Accuracy, Precision, Recall, F1	99.45% (CatBoost, URL+HTML+derived)
PhishOFE (Kustiawan & Ghauth, IEEE Access 2025)	PhishOFE dataset (own; Phish-Tank+OpenPhish phishing, Open PageRank legit)	101,063	48.97% (≈49,491)	51.03% (≈51,572)	URL + HTML + derived features; deliberately no third-party/WHOIS features	10 ML models compared; best = CatBoost	Accuracy, Precision, Recall, F1, MCC, train/predict time	99.48% (CatBoost)
Layered Model (Goenka et al., IEEE Access 2025)	Own DS1 (Phish-Tank+Kaggle) + 6 comparison sets (DS1_5k/10k/25k, DS2, DS3, DS4)	DS1 = 38,661	19,432	19,229	Layer 1: 5 brand-jacking features (fuzzy match); Layer 2: 19 lexical features	Layer 1: rule-based fuzzy matching; Layer 2: XGBoost (best of 5 ML models)	Accuracy, Precision, Recall, F1, response time (ms)	99.35% (12.49 ms); 89.17% on older DS2 (73,576 URLs)
X-PHIDE (Misiek & Hyla, IEEE Access 2026)	Custom (4 sources) + GramBeddings + PhiUSIIL	Custom 75,738; GramBeddings 639,723 train / 159,940 test; PhiUSIIL 235,795 (47,103 used)	PhiUSIIL: 100,945; Custom/GramBeddings: ≈50% (rate only)	PhiUSIIL: 134,850; Custom/GramBeddings: ≈50% (rate only)	>90 raw features reduced to 8 selected (length_url, qty_hyphen_path, has_brand_keyword, subdomain_auth_count, qty_nameservers, qty_dot_url, is_spf_domain, time_domain_activation)	XGBoost (Bayesian-tuned)	Accuracy, Precision, Recall, Specificity, F1, ROC-AUC, train/inference time, memory	90.72% in-distribution; 77.84% avg. cross-dataset
Staying Ahead of Phishers (Kavya & Sumathi, AI Review 2025) — <i>survey</i>	N/A — literature review; cites PhishTank, PhiKitA, UCI repo, Ethereum data across ~50 surveyed studies	N/A	N/A	N/A	Surveys list-based, ML, DL (CNN/RNN/LSTM/GRU), GAN, ensemble, NLP, and graph/network-embedding methods	N/A (reviews the field, not a single method)	Accuracy, Precision, Recall, F1, FPR, AUC-ROC, efficiency (as reported across cited works)	N/A — cited works range 91.5–98.9%
Uncovering the Cloak (Li et al., IEEE Access 2023) — <i>SLR</i>	N/A — systematic review of 30 primary studies, 2012–2022	N/A	N/A	N/A	Surveys client-side, server-side, URL-manipulation, and content-based cloaking, plus cloaking-detection techniques	N/A (reviews the field, not a single method)	N/A — one cited MITM-toolkit classifier reached 99.9%	N/A

Limitations and Research Gaps, per Paper

Paper	Limitation (as stated by the authors)	Gap / Future Work (as stated by the authors)
ResMLP	No dedicated limitations section; implicitly, the 7-block residual architecture adds complexity, and the authors state “achieving optimal detection accuracy remains an ongoing pursuit.”	Extend to multi-class classification; streamline/reduce the residual pipeline’s complexity; improve adversarial robustness; integrate real-time monitoring.
Feature Extension	URL-only: does not analyze page content or malicious JavaScript in phishing sites; BERT component is computationally expensive.	Add web-content detection for full coverage; optimize/lighten the BERT model; extend to blockchain and cryptocurrency-specific phishing threats.
Multimodal XAI	Misclassifies some popular legitimate sites due to coincidental structural similarity; screenshot model sensitive to resolution/aspect-ratio mismatches; late fusion builds no joint cross-modal representation; base models trained independently.	Move to a hybrid binding scheme with shared latent representations and cross-modal attention (esp. HTML+screenshot); joint cross-modal XAI; systematic adversarial-robustness evaluation.
Evaluating Feature Engineering	Not explicitly stated for their own study (limitations discussed are of prior/related work, not this paper’s).	Cascading architecture: cheap URL analysis first, heavier content/relationship analysis only for borderline cases; craft further derived features; ensemble methods with comprehensive feature sets.
PhishOFE	No dynamic JavaScript/interaction analysis (static features only); not designed for homograph attacks (Latin-script dataset only, not systematically evaluated against them); limited multilingual/internationalized URL coverage; performance tightly coupled to this specific dataset.	Homograph-attack detection via character normalization, visual-similarity metrics, and Unicode-aware features; unsupervised learning (clustering, autoencoders, one-class) for novel attacks; multilingual URL datasets; deployable end-user application.
Layered Model	Brand-jacking layer requires brand names of length ≥ 6 (shorter names risk false positives); restricted to a fixed list of most-phished brands, so squatting on unlisted brands relies on the lexical layer alone.	Expand the brand list; explore alternative transforms for length-based features (binning, rank, log-scale); apply deep learning; extend fuzzy matching to non-space/complex-morphology languages.
X-PHIDE	Accuracy drops substantially under dataset shift (70.65–90.72% range); large memory footprint limits browser deployment; no automatic retraining; not built for concurrent-request load; Chrome-only; vulnerable to evolving obfuscation (shorteners, homoglyphs, percent-encoding).	Model compression/distillation (pruning, quantization, ONNX); automatic model updating; multi-model ensembling; cross-browser support; adversarial retraining with runtime WHOIS/DNS validation checks.
Staying Ahead of Phishers (survey)	High false-positive rates in traditional ML/similarity methods; adversarial vulnerability; narrow/imbalanced datasets; poor cross-domain generalizability; “dark-box” deep-learning/GAN interpretability; high compute cost; concept drift as attacks evolve.	Explainable AI (saliency maps, attention, LIME); federated learning/differential privacy; contextual/behavioral analysis; cross-domain transfer learning; adaptive/incremental retraining; model compression (pruning, quantization, distillation); stronger adversarial defenses.
Uncovering the Cloak (SLR)	Most detection is server-based, sharing blacklists’ latency weakness; very few detection strategies account for cloaking at all; a small number of sources drive over 80% of successful attacks; crawler ecosystems lack fingerprinting diversity.	Broaden temporal scope and databases; diversify literature types reviewed; conduct focused studies of individual cloaking/detection mechanisms; target the small set of sources behind most attacks; improve anti-phishing data-sharing and takedown collaboration.

Note: two papers (Kavya & Sumathi; Li et al.) are literature reviews rather than original experiments, and are marked N/A where a column requires a primary dataset, algorithm, or accuracy figure that a review does not produce on its own.